

Esthetic Inlays and Onlays

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There are several treatment options for conservative posterior esthetic restorations in addition to direct resin composite restorations. This chapter discusses tooth-colored inlays and onlays fabricated in resin composite and in ceramic materials. Restorations fabricated with computer-aided design/computer-assisted manufacture (CAD/CAM) technology will also be discussed. Esthetic inlays and onlays have a number of characteristics in common, whether they are resin, ceramic, or fabricated with CAD/CAM technology.

General Considerations

Preparations

The preparations for ceramic and resin composite inlays and onlays are essentially the same. Preparations for CAD/CAM inlays and onlays do not differ from preparations for laboratory-fabricated inlays and onlays. There is little research to

support the efficacy of any preparation design over another.[1] However, based on knowledge of the materials and clinical experience, the divergent, relatively nonretentive preparation is most commonly advocated because of ease of placement (Fig 19-1). Resistance form may be incorporated with rounded proximal boxes, but grooves should not be used. Resistance and retention form for the restoration are provided primarily by adhesion to enamel and dentin. Supragingival margin placement is desired to facilitate isolation for the adhesive cementation of the inlay or onlay. The walls and floors of the preparation should be smooth and even, and the internal angles should be rounded to enhance adaptation of the restorative material. Occlusal reduction should be uniform and of sufficient thickness to ensure strength of the overlying ceramic material to avoid fracture. This equates to at least 1.5 mm in the central fossa and 2.0 mm over functional cusps[2] (Fig 19-2).

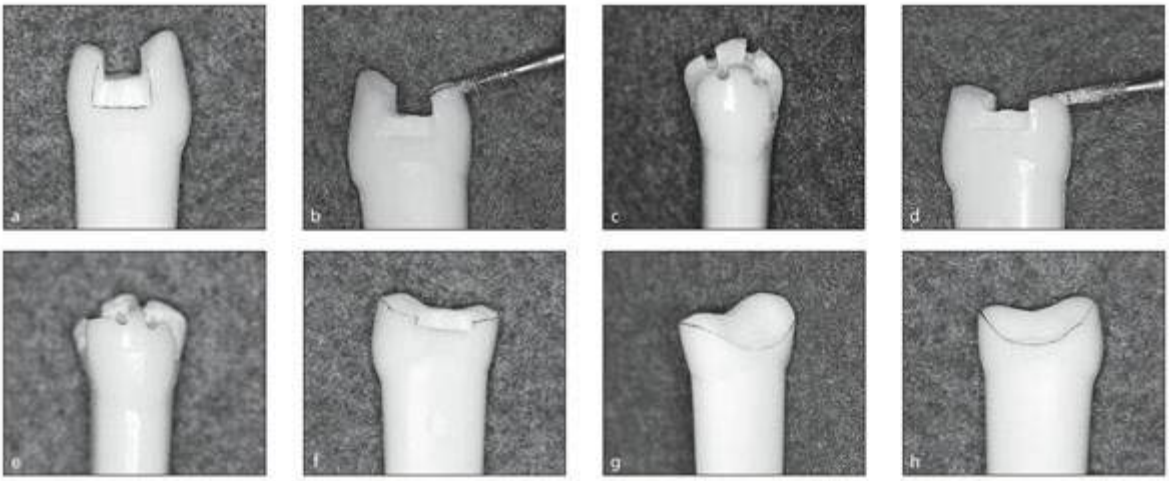


Fig 19-1 Onlay preparation technique. *(a)* Simulation of amalgam preparation after removal of the amalgam. *(b)* Use of a diamond of known diameter to make the first depth cut of 1 mm. *(c)* First depth cuts of 1 mm. *(d)* Use of a diamond of known diameter to make the second depth cuts of 1 mm. *(e)* Second depth cuts of an additional 1 mm. *(f)* Occlusal reduction of 2 mm completed; however, sharp line angles remain in the box area. *(g and h)* Completed smooth, flowing preparation without sharp angles.



Fig 19-2 Inadequate occlusal reduction for a porcelain onlay, resulting in fractured porcelain.

There is no benefit to the placement of bevels at the occlusal or gingival margins; in fact, bevels should be avoided because thin margins of both resin composite and most ceramics are susceptible to chipping during function.[3,4] A 90-degree butt joint minimizes the chipping problem but may result in a visible demarcation between the tooth and the restoration. Therefore, when the esthetic blend of the restoration and the tooth is important, such as on the facial surface of a maxillary premolar, the entire facial surface of the premolar may be included in the preparation to ensure an optimal shade match with the adjacent teeth (Fig 19-3). A more recent preparation modification has been suggested to enhance the esthetic blend of the facial margin and not involve the entire facial surface of the tooth in the preparation. The cavosurface margin is modified with a football-shaped diamond at a 45-degree angle to the cavosurface margin, creating a blend of porcelain to enamel shading and still maintaining a bulk of ceramic at the margin for physical strength of the ceramic. An in vitro study on ceramic preparations for conservative restoration of endodontically treated teeth reported a significant improvement in the marginal and internal fit of the ceramic restoration with this modified-margin design relative to both a ferrule and a 90-degree butt-joint margin.[5]

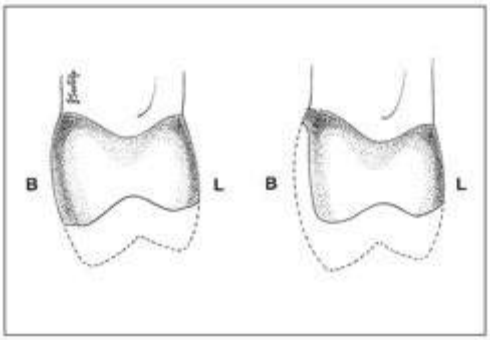


Fig 19-3 Standard onlay preparation *(left)* and modified onlay preparation *(right)* that includes coverage of facial surface to achieve a superior esthetic blend with the natural tooth color. B—buccal; L—lingual.

Bases and liners

The use of bases and liners is somewhat controversial. Initially, glass-ionomer bases were used for dentinal protection and to base the preparation to "ideal" form. However, it has been shown that it is not necessary to protect the dentin from the phosphoric acid etchant.[6] Therefore, glass-ionomer cement is recommended only for routine block-out of undercuts.

Provisional restorations

Provisional restorations can be a challenge with esthetic inlays and, particularly, onlays because of the nonretentive design of the preparations. Provisional restorations can be made in the usual manner with acrylic resin or resin composite and cemented with temporary cement. It is commonly stated that a eugenol-containing cement should not be used with the provisional restoration when the definitive restoration will be bonded with a resin cement.[1,7] However, the literature is equivocal on the deleterious effect of eugenol. Some studies support the hypothesis that a eugenol-containing provisional cement inhibits the set of resin cements,[1,7,8] while others show that eugenol-containing provisional cements have no impact on bond strengths.[9–14] Because of the nonretentive design of the onlay preparation, the more retentive polycarboxylate cement is the temporary luting cement of choice. Small mechanical undercuts should be placed on the intaglio surface of the provisional restoration to aid in retention. If adjacent teeth are being restored, the provisional restorations can be connected to improve retention. Alternatively, a local internal area of the preparation away from any margins may be "spot etched" with phosphoric acid to adhesively bond the provisional restoration in place. A finishing diamond bur may be required to remove the area of bonding once the provisional restoration has been removed to ensure complete removal of the adhesive from the preparation surface prior to delivering the restoration.

For inlay preparations, semiflexible light-cured materials, such as Systemp (Ivoclar Vivadent) or Barricade (Dentsply) may be used. To provide retention and to decrease sensitivity, a dentin primer can be placed in the preparation and air dried, or a small amount of resin-modified glass-ionomer cement liner can be placed on the pulpal floor and polymerized. The preparation is filled with the provisional material, and the patient is instructed to bite into maximum intercuspation to develop the occlusion. Excess material is removed with an explorer, and the provisional restoration is light cured. This technique is recommended only for short-term use in small preparations.[15]

Adhesive cementation

Resin luting cement is the only material recommended for cementing this type of restoration because it bonds to enamel, dentin, and the restorative material. Luting resin cement limits microleakage and enhances the strength of the restoration.[16] It is commonly reported that indirect bonded restorations strengthen the prepared tooth[17,18]; however, this is not universally supported by the literature.[19] Dual-cured luting resin and dual-cured dentin bonding agents, which combine light-curing and

chemical-curing components, should be used to bond all indirect posterior bonded restorations.[20,21] The light-curing component polymerizes rapidly on exposure to light of the proper wavelength, while the chemical-curing component undergoes a slow polymerization process in those areas to which the light does not penetrate. It is important that the curing light be applied to dual-cured resin for an adequate period of time, because the dual-curing process results in more complete polymerization than that achieved with chemical polymerization alone.[22,23] The shelf life of dual-cured resins is shorter than that of conventional light-cured resin composites. Therefore, expiration dates of adhesive agents and resin luting cements should be monitored carefully. Alternatively, a test batch of dual-cured luting resin can periodically be mixed and allowed to cure in a dark environment to ensure that it will polymerize in the absence of light. Most dual-cured luting resins should polymerize at room temperature in the dark within 10 minutes.

Preparing the restoration for bonding

Adhesion is more reliably achieved to ceramic materials than to resin composite restorations. Ceramics used in these restorations must be etched, creating durable micromechanical retention.[24] The ceramic inlay/onlay is prepared for bonding by etching the internal surface, usually with hydrofluoric acid. This is generally done at the laboratory but may be done chairside instead. Shortly before bonding, a silane coupling agent is applied to the etched surface to enhance wetting by the resin adhesive.[25,26]

Bonding to resin composite restorations is more difficult. The intaglio surface has no air-inhibited layer and relatively few unreacted methacrylate groups, so a reliable chemical bond does not form between the inlay and the resin cement.[27–29] Because the resin composite–cement interface may be the weak link, several procedures have been recommended to enhance the bond to resin. With hybrid resin composite, the intaglio surface should be carefully air abraded with 50-µm aluminum oxide or 30-µm silica-modified corundum particles (CoJet Sand, 3M ESPE), avoiding the margins[30] and then cleaned with a steam cleaner or in an ultrasonic bath. The air abrasion provides a rough surface for frictional retention. The cleaned surface should be treated with an agent to allow better wetting by the cement. Silane is generally the preferred wetting agent.[28,31,32] While treatment with hydrofluoric acid has also been recommended to etch the glass particles in the hybrid resin composite, laboratory research does not support the efficacy of this procedure.[27]

Preparing the tooth for bonding

Ensuring an isolated operating field is a critical step in the adhesive bonding process. Only in a controlled environment can the adhesive bonding materials achieve maximum strength. There are several useful devices. The most cost-effective device is the rubber dam (Fig 19-4a). A number of other commercially available devices, such as an OptraGate (Ivoclar Vivadent) or Isolite[2] (Isolite) (see chapter 8), may be employed if the use of a rubber dam is not feasible or desirable for the clinical case at hand.

Once the restoration is ready for bonding, a decision must be made regarding the type of dentin bonding agent. Three-step etch-and-rinse (fourth-generation) light-cured-only adhesive systems should not be used under indirect posterior bonded restorations. The light-cured adhesive would need to be air thinned prior to polymerization to ensure the absence of pooling of the adhesive, which would prevent complete seating of the restoration. Moreover, it has been shown that air thinning of the light-cured adhesive resin significantly decreases the bond strength.[33] This leaves three-step etch-and-rinse (fourth-generation) dual-cured systems as the adhesive system of choice. However, any adhesive agent that will cure in conjunction with a dual-cured resin luting agent can be selected, regardless of the type of etching system or the number of application steps. Adhesive systems are described in detail in chapter 9.

The technique with the most research support employs the dual-cured, three-step etch-and-rinse (fourth-generation) dentin bonding systems. The enamel is etched for 30 seconds and the dentin for 15 seconds, and the tooth is rinsed for 15 seconds and air dried to enable visual inspection for an adequate enamel etch. The dentin is remoistened with chlorhexidine, and several coats of a hydrophilic resin primer are applied to infiltrate the exposed collagen fibers and form the hybrid layer. Active application of the primer on the dentin may enhance bonding,[34] but the specific instructions of the manufacturer should be followed carefully because they vary from brand to brand. The primer is completely air dried with gentle air pressure. The dual-cured adhesive is placed on the tooth and the previously etched and silanated intaglio surface of the restoration. The dual-cured luting resin is mixed and placed into the preparation and restoration, and the restoration is seated. Excess resin must be completely removed from the proximal-surface gingival margins with floss and interproximal instruments before polymerization of the luting resin. Some clinicians recommend placement of a clear gel, such as glycerin, on the margins before polymerization to prevent the formation of an air-inhibited layer in the luting resin at

the margin.[35] The margins may be finished with microfine diamonds or multifluted carbide finishing burs and a no. 12 or 12B scalpel blade, and they can be polished with disks, rubber points, or cups. After removal of the rubber dam, the occlusion is adjusted and the surface is polished to a high shine. The final step is rebonding, as described in chapter 11, with an unfilled or lightly filled resin (Figs 19-4b to 19-4g). Figure 19-5 demonstrates luted restorations.



Fig 19-4*(a)* Rubber dam isolation for cementation of a quadrant of porcelain onlays. *(b)* Master cast of a quadrant of porcelain onlay preparations. Note the amount of occlusal reduction. *(c)* Occlusal view of onlay preparations. *(d and e)* Porcelain onlays 2½ years after placement. (Porcelain onlays created by Gilbert Young, CDT, Plano, Texas.) *(f)* Lateral view of the maxillary and mandibular quadrants of porcelain onlays. The esthetic blend on the facial aspect of the maxillary premolars is better because the preparations were taken further gingivally than on the molars. *(g)* Bitewing radiograph taken after 2½ years showing porcelain onlays. Note the different cement radiopacities in the maxillary and mandibular restorations.



Fig 19-5*(a)* Porcelain onlay preparation, man-dibular first molar. *(b)* Master cast of the prepared mandibular first molar for a porcelain onlay. *(c)* Completed porcelain onlay on the mandibular first molar. *(d)* Buccal view of the completed porcelain onlay on the mandibular first molar. (Porcelain onlay created by Gilbert Young, CDT, Plano, Texas.)

Despite the initial high bond strength of two-step self-etching (sixth-generation) systems and one-step self-etching (seventh-generation) all-in-one systems, these values have been shown to decrease significantly under thermal and mechanical stresses, particularly on enamel; therefore, etch-and-rinse adhesives continue to be the preferred system for dental adhesion.[36,37] Alternative adhesive cementation techniques may be considered by combining a different generation of bonding agent with a dual-cured resin cement. Generally, this results in fewer clinical steps required for the adhesive bonding. However, this is not without a degree of risk. Many two-step etch-and-rinse (fifth-generation) and one-step self-etching (seventh-generation) products are incompatible with dual-cured and autocured luting composites.[38] The clinician must ensure that the selected bonding agent and resin cement will polymerize together.

Maintenance

Maintenance is a very important factor in the longevity of esthetic inlays and onlays. As with all types of restorative dentistry, lack of preventive measures and the resulting caries can cause failure of the finest restoration. Use of devices such as ultrasonic scalers or air-abrasive polishers on these restorations must be avoided because they can cause surface and marginal damage. Calculus should be removed carefully with hand instruments. When scalers are used around a bonded inlay or onlay, care must be taken not to chip the margins. Surface stain may be removed

from a restoration with aluminum oxide polishing paste or diamond polishing paste on a felt wheel or rubber cup. Because of their ability to etch porcelain, neither acidulated phosphate fluoride nor stannous fluoride solutions or gels should be used intraorally in patients with ceramic restorations. Only neutral sodium fluoride solutions should be used.

The patient should be advised that foods and liquids with a high potential for staining, such as coffee and tea, increase the potential for marginal staining. The patient must also be made aware of the potential for restoration fracture. Activities such as ice chewing and nail biting absolutely must be avoided. When a patient has a history of a parafunctional habit, a protective appliance should be fabricated to protect both the inlay or onlay and the opposing teeth.

Resin Composite Inlays and Onlays

Inlays and onlays made of resin composite are quite popular in Europe but have not gained wide acceptance in the United States. These restorations are typically fabricated on a gypsum cast or die. After polymerization, the restoration is bonded in place with a resin luting cement. Resin composite inlays can be highly esthetic and have certain advantages over direct resin composite and bonded ceramic restorations.

Advantages over direct resin composite restorations

As discussed in chapter 11, inadequate proximal contours and open contacts can be common problems of direct resin composite restorations. These are fewer common problems with resin inlays because contours and contacts can be developed outside of the mouth. If a contact is inadequate, it can easily be corrected prior to cementation.

Several problems associated with direct resin composite restorations are the result of polymerization shrinkage. During polymerization, resin composite shrinks on the order of 2% to 4%,[39–41] often causing a gap to form at the least retentive marginal interface, which is usually the gingival margin. Microleakage and bacterial ingress into the marginal gap may cause pulpal irritation and tooth sensitivity.[42] Current dentin adhesives have lessened, but not eliminated, the problem.[43,44] Polymerization shrinkage can also cause cuspal flexure, which is sometimes associated with craze lines in the enamel and postoperative sensitivity.[45]

In theory, polymerization shrinkage should be less of a problem with resin inlays because they are polymerized before cementation. The only polymerization shrinkage that occurs at the time of cementation is in the thin layer of resin cement, though this can be considerable (ie, 2% to 7%).[40,41] In fact, polymerization shrinkage stress in thin resin cement layers can be as great as or even higher than in thicker composite increments.[46] Resin inlays are reported to have less microleakage[47–49] and greater strength and hardness[50–52] and result in less postoperative sensitivity than direct resin composite restorations.[53]

While the in vitro data suggest that there are significant advantages to the resin inlay compared with the direct resin composite restoration, this is not corroborated by the in vivo data. In a 5-year clinical study[54] and another 11-year clinical study,[55] no statistically significant difference was found in the success rates for resin inlays compared with direct-placement resin composite restorations. Based on these data, and because of the increased cost of the resin inlay and the frequent need to remove more solid tooth structure for inlays, it is currently difficult to make a strong case in favor of the resin inlay over the direct resin composite restoration.

Secondary polymerization

The superior physical and mechanical properties of resin inlays over direct composite restorations are primarily due to more complete polymerization resulting from secondary polymerization procedures. Directly placed resin composites harden through a process of free radical polymerization of methacrylate groups. In most cases, the polymerization reaction is initiated when a molecule within the resin composite (camphorquinone) forms free radicals when exposed to light of the appropriate wavelength (about 470 nm). The radicals react with a photoreducer (an aromatic or aliphatic amine) to initiate chain formation of the methacrylate groups. As polymerization progresses, the methacrylate chains grow and the material loses its fluidity as a highly cross-linked, rigid network polymer forms. Even with long curing times and powerful lights, incomplete polymerization occurs, particularly at greater depths below the surface because of inadequate light penetration and the immobilization of unreacted molecules within the cross-linked polymer network.[56]

Light-cured resin composite inlays undergo this initial polymerization but then are further polymerized with a combination of intense light, heat, and/or pressure. The postcure can be performed in a postcuring unit specifically made for this purpose, in a toaster oven at approximately 250°F for 7 minutes,[50] or with a curing light or light box. Polymerization under pressure has been shown to increase both diametral

tensile strength and stiffness of the restoration.[57] These secondary curing procedures are recommended with all indirect resin systems, although they may not be mentioned in all manufacturers' instructions.

Posterior Bonded Ceramic Restorations

Ceramic inlays were introduced in 191358 but did not become popular because of difficulties in fabrication and a high failure rate.[1] In the 1980s, the development of compatible refractory materials made fabrication easier, and the development of adhesive resin cements greatly improved clinical success rates.[2]

The modern generation of bonded porcelain restorations was first described in 1983.[59] When it became clear that the technique had merit in anterior applications, interest developed in the use of bonded porcelain for posterior applications. In 1986, Redford and Jensen[60] described the strengthening effect of porcelain inlays on the fracture resistance of natural teeth. In 1988, Jensen[61] reported excellent clinical success in a 2-year in vivo study. The technique has since been refined to the point that porcelain inlays and onlays are now an accepted operative modality. Box 19-1 details the indirect tooth-colored restoration technique.

Box 19-1	**Indirect tooth-colored onlay technique**
Tooth preparation 1. Select a shade prior to tooth dehydration. 2. Make a stent for fabrication of a provisional restoration. 3. Remove caries and any existing restorative materials. 4. Preparation should be well isolated to ensure success with the adhesive cementation technique. 5. Preparation should allow 2 mm of occlusal clearance for the definitive restoration. All internal line angles should be rounded and walls divergent occlusally. There should be no grooves or sharp angles. 6. When necessary, retract gingival tissues to expose the preparation margins. Take a final impression. 7. Make a custom provisional restoration	17. Remoisten the dentin preferably with chlorhexidine and actively apply several coats of dentin primer on damp dentin. Air dry the primer, gently at first, until the surface is completely dry, and confirm a uniform shiny surface. 18. Remove the matrix or Teflon tape. 19. Mix and place a dual-curing adhesive, which should not be light cured before placement of the restoration; similarly, dual-curing adhesive must be placed on the inner surface of the restoration. 20. Mix the dual-curing resin composite luting cement and

using the stent.
Place undercuts in the intaglio surface of the provisional restoration.
8. Cement with a strong provisional cement; because the preparation has minimal resistance form, polycarboxylate cement is the cement of choice for luting the provisional restoration.

Restoration placement
9. Check the restoration on the die for fit, and check for fracture lines with transillumination.
10. Isolate tooth preparation to prevent contamination during the adhesive bonding sequence.
11. Try in the restoration; adjust contours and contacts with appropriate polishing disks and points.
12. Clean the onlay with acetone and air dry.
13. Etch the internal surface of the ceramic restoration with hydrofluoric acid, rinse thoroughly, and air dry. Air abrade the inner surface of the composite inlay with 50-μm alumina or 30-μm silica-modified corundum particles (CoJet Sand).
14. Place silane on the etched internal surface of the onlay and air dry.
15. A matrix band or Teflon tape can be used to protect the adjacent teeth.
16. Etch the tooth with 30% to 40% phosphoric acid etchant gel for 15 to 20 seconds, wash for 5 seconds, and air dry to ensure an adequate enamel etch.
place into the preparation and the inner surface of the restoration with a syringe.
21. Gently place the restoration into the preparation and vibrate with a hand instrument to ensure that it is almost fully seated.
22. Remove excess resin composite luting cement with a brush both occlusally and interproximally.
23. Gently seat the restoration completely with an instrument applied to the occlusal surface, making sure that a bead of composite is expressed at all margins. Confirm correct seating with an explorer at the margins.
24. While the assistant holds the restoration in place, gently clean the interproximal margins with floss, an explorer, and a no. 12 or 12B scalpel blade, being careful not to cause bleeding. The interproximal margins must be completely finished before the resin composite polymerizes. Leave excess composite luting cement on the facial and lingual margins.
25. Cover all accessible interproximal margins with glycerin or with water-soluble lubricant.
26. Complete polymerization by light curing for 90 seconds from the occlusal aspect and 30 seconds each from the facial and lingual aspects in interproximal areas.

Finishing
27. Finish all margins with 12-fluted carbide burs or microfine diamonds, finishing disks, and/or composite polishing points.
28. Adjust the occlusion with articulating paper and a microfine diamond.
29. Complete polishing with appropriate ceramic or composite polishing points.

Indications

The indications for posterior bonded porcelain restorations overlap those for direct

and indirect posterior resin composite restorations, which have already been described. These restorations are indicated when there is an overriding desire for esthetics and all margins can be placed on enamel. Some clinicians have recommended bonded porcelain rather than indirect resin composite for larger restorations.[2]

Ceramic inlay versus resin composite inlay

Ceramic inlays are reported to leak less than resin composite inlays.[62–64] The marginal fidelity depends on technique and is laboratory dependent.[62] One laboratory study demonstrated better marginal fidelity with ceramic inlays,[64] while another study found better fit with resin inlays.[65] Pressed ceramic inlays have been shown to have an average marginal gap of less than 50 μm.[66] As previously described, adhesion of luting resin is more reliable and durable to etched porcelain material than to prepolymerized resin composite. Therefore, the main advantage of ceramic over composite inlays relates directly to the reliability of the cementation process.

The main advantage of resin composite inlays is that they tend to be more user-friendly, both clinically and in the dental laboratory. The resin inlay can be placed into the preparation with moderate pressure to ensure complete seating without the fear of fracture. Also, if the interproximal contact is removed during the process of adjustment, it can easily be replaced with the addition of light-cured resin composite.

In contrast, the ceramic inlay is quite fragile and subject to fracture during the try-in phase. If the interproximal contact is inadvertently removed during adjustment, it can be replaced by adding low-fusing porcelain, but this step requires refiring the inlay in a porcelain furnace.[67] Once it is removed from the porcelain furnace, the inlay must be re-etched with hydrofluoric acid prior to cementation.

One in vivo study[68] compared the clinical success of these two treatment modalities at 3 years and found that Class 2 ceramic inlays had a significantly higher breakage rate than indirect resin composite inlays. This finding was corroborated in a laboratory study[69] that found lower fracture resistance of ceramic inlays compared with resin inlays. Although a direct composite restoration is recommended over an inlay, as stated previously, there are instances when an inlay could be well recommended. Such situations include a large interproximal distance between teeth leading to difficulty in achieving proper interproximal contact, the need to create a specific occlusal relationship, and defects that extend more than two-thirds of the

intercuspal width (ie, large isthmus). However, based on the limited data available, resin composite should be considered first for use in two- and three-surface inlay preparations of moderate width.

Ceramic onlay versus resin composite onlay

The ceramic onlay has the same disadvantages as the ceramic inlay. Although some ceramic materials cause wear of opposing enamel,[70] they also provide long-term occlusal stability, which resin composite may not provide in a cuspal-coverage restoration.[71] The stronger bond of resin cement to porcelain is particularly important when cusps are covered. The stronger the bond, the more efficiently forces are transferred through the restoration and the cement and absorbed into the tooth.[16] For these reasons, when even one cusp of a posterior tooth is being covered with an esthetic bonded onlay, the ceramic onlay is preferred.

Ceramic onlays may be used routinely for the esthetic restoration of premolars. They may also be used as cuspal-coverage restorations for molars, although the occlusal forces will be greater in the molar region. Another indication for the ceramic onlay is in the restoration of a molar with a short occlusogingival dimension. In this circumstance, it is difficult to gain axial retention and resistance with a conventional crown preparation. However, the porcelain onlay preparation requires only 2 mm of occlusal reduction and no axial reduction for retention and resistance. The short molar, which would have previously required crown lengthening surgery before placement of a complete-coverage crown, can now be restored more conservatively with the porcelain onlay.

Selection of appropriate patients is of paramount importance in the placement of posterior bonded porcelain restorations. For the greatest long-term predictable success, all margins should be on enamel because of the more reliable bond and seal as compared with dentin. Also, the patient and the tooth to be restored should be amenable to predictably good tissue retraction and isolation from moisture to provide the best environment for bonding the restoration. Ideally, the patient should exhibit no signs of a parafunctional habit. In addition, the restoration should be fabricated so that it contacts in maximum intercuspation position of the mandible but has no contact on the porcelain in mandibular excursive movements.

Shade selection

The shade used for a ceramic inlay or onlay is selected in the same way as for a metal-ceramic crown. Because of the thickness of the occlusal porcelain, the

underlying tooth color and cement shade have a minimal effect on the shade of the definitive restoration except at the margins. As with porcelain veneers, use of a translucent resin cement is recommended to improve the esthetic blend at the margins.

Fabrication

The most common method of fabrication of ceramic inlays and onlays utilizes a refractory die. After a master die is poured in die stone, a refractory die is made by duplicating the master die or repouring the impression in a refractory material. The porcelain is baked on the refractory die, recovered, and fit to the master die. Variations in the fit of ceramic inlays and onlays are reported to be related more to the ability of the technician than to the type of ceramic material used.[66]

The newer generation of pressed ceramics is fabricated much differently. The restoration is waxed on a stone die in the traditional manner and invested in a special investment. The invested wax pattern is burned out as in the traditional lost-wax technique. An ingot of the pressed ceramic material is heated and pressed into the lost-wax pattern space. After cooling, the investment is removed and the ceramic restoration is retrieved and finished in the same manner as a feldspathic porcelain restoration. Another very popular method for fabricating indirect ceramic restorations is via CAD/CAM systems, which are described in detail later in the chapter.

Isolation

It is universally acknowledged that strict isolation is necessary for bonding posterior adhesive restorations. This is best accomplished with a well-placed rubber dam; however, other techniques can also be used to prevent contamination during the adhesive bonding sequence. If it is not possible to isolate the tooth, an adhesive restoration should not be placed (see Box 19-1).

Resin Composite versus Ceramic

Wear

There are significant differences in the wear characteristics of resin composite and ceramic. Wear is not a significant factor in a ceramic restoration,[2] but traditional

feldspathic porcelain is highly abrasive and can cause significant wear of the opposing dentition. A newer generation of low-fusing porcelains has been shown to cause significantly less wear of enamel than traditional feldspathic porcelain.[72,73] The new class of pressable ceramics, described previously, has become popular, in part because they are less abrasive to opposing teeth. These include Empress (Ivoclar Vivadent), Empress 2, which has been replaced with IPS e.max (Ivoclar Vivadent), Optimal Pressed Ceramic (OPC, Jeneric/Pentron), and Finesse All-Ceramic (Dentsply). The first generation of Empress showed decreased wear of opposing enamel compared with traditional and low-fusing porcelains in vitro.[74] However, the newer generation, Empress 2, has shown almost no wear of opposing enamel, both in vitro[73] and in vivo at 6 months.[75] If long-term clinical studies confirm the low wear of opposing enamel, this will be a significant advance in ceramics.

The data concerning wear of resin composite materials have been somewhat contradictory.[76–78] Enamel is reported to wear at a rate of 30 μm per year in molars and 15 μm per year in premolars.[79] Most modern resin composite materials fall within that range.[78,80] Ferracane[80] reported that the wear of the traditional resin composite, Charisma (Heraeus Kulzer), was lower than that of the indirect composite, Artglass (Heraeus Kulzer). Similarly, Reich et al[81] evaluated the wear of indirect resin composites, Artglass and Targis (Ivoclar Vivadent), to a traditional resin composite, Z100 (3M ESPE). Although Targis demonstrated superior physical and mechanical properties, Z100 had the highest wear resistance. In another laboratory study,[82] the wear of three commercially available indirect resin composite materials was compared with a cast gold control. Targis demonstrated the greatest wear, followed by Artglass; Skulptur FibreKor (Jeneric/Pentron) demonstrated the least wear, which was approximately equal to that of gold. Similarly, the wear rates for Artglass and Targis were evaluated in a two-body wear test.[83] Targis demonstrated wear similar to that of enamel, and Artglass had significantly higher wear than enamel. Another laboratory study[84] reported that the wear resistance of indirect resin composites is similar to that of gold but is significantly more abrasive to antagonistic enamel.

Longevity

Results of short-term clinical studies of composite inlays are encouraging, but there is little long-term data. Bishop[85] reported one failure out of 92 composite inlays that

had been in place for 7 months to 4 years. A Swedish study[86] reported that 29 of 30 resin inlays were excellent or acceptable at 17 months, while another study[87] reported good marginal integrity at 5 years. Wendt and Leinfelder[88] reported no failures among 60 resin inlays after 3 years, while another study[68] reported that 10 of 145 composite inlays failed at 3 years. A prospective clinical study by Leirskar et al[89] evaluated 64 indirect composite inlays/ onlays over a period of 48 to 75 months with a mean time of 59 months. They reported three failures (5%) because of fractures (two) and caries (one). However, 18 of 64 restorations were rated as less than optimal, mostly because of the change of interproximal contacts over time.

It is still not clear whether the resin inlay offers any advantages in terms of longevity over the direct-placement resin composite restoration. Two clinical studies reported no significant differences between composite inlays and direct resin composite restorations at 5 years[54] and at 11 years.[55] One clinical study[90] of belleGlass (Kerr) reported 5-year results for 24 inlays and onlays. All restorations performed satisfactorily, although 12% had interfacial staining and 58% had slight to moderate marginal degradation. In contrast, another clinical study[91] evaluated 99 composite inlays over a period of 6 to 53 months and reported a success rate of 98%. A recent 36-month clinical study showed 100% success of molar onlays fabricated from two indirect resin composites (Admira and Grandio, Voco), although there was some marginal degradation over the course of the study.[92]

Several clinical studies have evaluated the performance of ceramic inlays or onlays. The clinical success of traditional feldspathic ceramic inlays or onlays has been mixed. A 6-year retrospective study[93] reported a 25.8% failure rate among feld-spathic porcelain onlays with and without metal reinforcement. One 4-year study[94] reported no failures in 50 inlays. Another study reported a 95% success rate after a mean evaluation period of 5.9 years.[95] Other clinical studies, however, have shown significantly higher failure rates. Molin and Karlsson[96] reported that 21 of 145 inlays fractured at 3 years, and their results were comparable with a study by Qualtrough and Wilson[97] that reported 16% failure at 3 years. Still another study[98] reported a 20% failure rate at 8 years, while a similar study[99] reported an 84% success rate with a mean observation time of 6.3 years. If these high failure rates are confirmed by other clinical studies, the long-term prognosis and indication for feldspathic porcelain inlays appear to be questionable.

Clinical results with the leucite-reinforced pressed ceramic restorations have been more promising. Several studies[100–103] have shown excellent clinical success up to 7 years. However, a randomized 5-year clinical evaluation[104] of three ceramic

inlay systems and gold inlays reported less-promising results: 20% of the leucite-reinforced ceramic inlays failed, and 70% of all of the ceramic inlay systems demonstrated marginal ditching, while there were no failures among the gold inlays. A 3-year study of lithium disilicate (IPS Empress 2) inlays showed 100% success of all 33 restorations available for recall.[105] Krämer and Frankenberger[106] evaluated the clinical outcomes of IPS Empress inlays as well as onlays with cuspal replacement, all with margins that were below the cementoenamel junction, and reported only 8% failures of 96 restorations in 8 years. Stoll et al[107] reported over 96% success over 10 years for a large number (N = 1624) of IPS Empress inlays and partial crowns, with the results being better for vital versus nonvital teeth. A relatively recent review of the clinical success of ceramic restorations for a variety of applications suggests that these materials are very effective.[108]

Failures

Two types of failure are most common with esthetic inlays and onlays: *(1)* bulk fracture (see Fig 19-2) and *(2)* marginal breakdown (Fig 19-6). Bulk fracture sometimes occurs in areas of cus-pal coverage, particularly if the restorative material is less than 2 mm thick (Fig 19-7). It also occurs at the isthmus adjacent to a marginal ridge, where the porcelain is poorly supported by tooth structure.



Fig 19-6 Maxillary first molar with a 7-year-old bonded Dicor (Dentsply) inlay with marginal ditching and a small fracture of the marginal ridge.

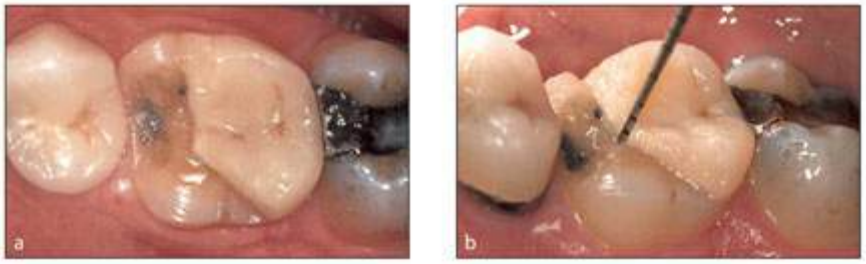


Fig 19-7*(a and b)* Maxillary first molar with a 5-year-old bonded feldspathic porcelain onlay demonstrating bulk fracture.

Marginal wear is a common finding in esthetic inlays and onlays.[90,104–111] Because resin cements tend not to be heavily filled, they wear more quickly than the adjacent restorations or tooth structure. This is particularly true if the marginal fit is poor.[112,113] Kawai et al[114] demonstrated a linear relationship between wear of resin cement and the horizontal marginal gap. They concluded that reduction of the marginal gap is an important clinical consideration in minimizing the wear of the resin cement. They also found hybrid resin cements to wear faster than microfilled resin cements. Isenberg et al[115] reported 3-year results of a clinical study of 121 CEREC (Sirona Dental) inlay and onlay restorations. None of the restorations exhibited any evidence of interfacial staining, discoloration, or caries, but about 50% of the restorations exhibited gap dimensions large enough to be detected with an explorer. The rate of wear of the resin composite luting agent was linear over the first year, but no further cement wear was noted over the course of the investigation. The depth-to-width ratio of the gap generally did not exceed 50%. However, Hayashi et al[116] reported results of an 8-year clinical study that evaluated marginal wear using an optical laser scanner. They found that marginal deterioration occurred in a sequential three-stage pattern. The initial stage of marginal deterioration occurred rapidly from initial placement to 21 months. In the second stage, the degree of deterioration leveled off and progressed at a much slower rate. The duration of this stage was 21 to 72 months. During the third stage, from year 6 to year 8, the marginal deterioration again accelerated to a much faster rate.

Computerized Dentistry

Computerized dentistry and *digital dentistry* are terms generally used to describe the clinical application of CAD/CAM techniques. There are three features common to CAD/CAM systems.[117] The first is the ability to record the patient's intraoral condition to the computer. This usually involves some sort of intraoral camera to capture a digital file of the dentition. Once the digital file is recorded, a software program is used to design the desired contours of the restoration. This involves control of the usual restoration parameters, such as proximal contacts, emergence profile, and occlusal contacts. Once the design has been completed, a machining device is used to produce the designed restoration. Most commonly, the machining device is a subtractive milling chamber that cuts the definitive restoration from a preformed block of restorative material.

Commercially available computerized dentistry systems that are used in the dental

office are generally considered to have two applications: *(1)* as a digital impression system and *(2)* as a chairside CAD/CAM system. Digital impression systems focus on the first step of the CAD/CAM process, an accurate recording of the patient's intraoral condition. A data file is created on the system computer, and the patient's dentition is scanned or recorded to the data file. The digital data file is then electronically transmitted via the Internet to a dental laboratory. The dental laboratory has two options for using the patient's data file. One option is to send the electronic data to a processing center to have models fabricated and mounted on an articulator. The articulated models are returned to the dental laboratory, and the restoration can be fabricated by any conventional laboratory process, such as those used for cast gold or porcelain-fused-to-metal restorations. The laboratory also has the option of using a software CAD program to design and produce either a coping or full-contour restoration. The articulated models are used to finish the case, as is done for zirconia or similar restorations[118] (Fig 19-8). The Lava Chairside Oral Scanner (COS) (3M ESPE) and iTero system (Cadent) are the two most common digital impression systems (Figs 19-9 and 19-10).

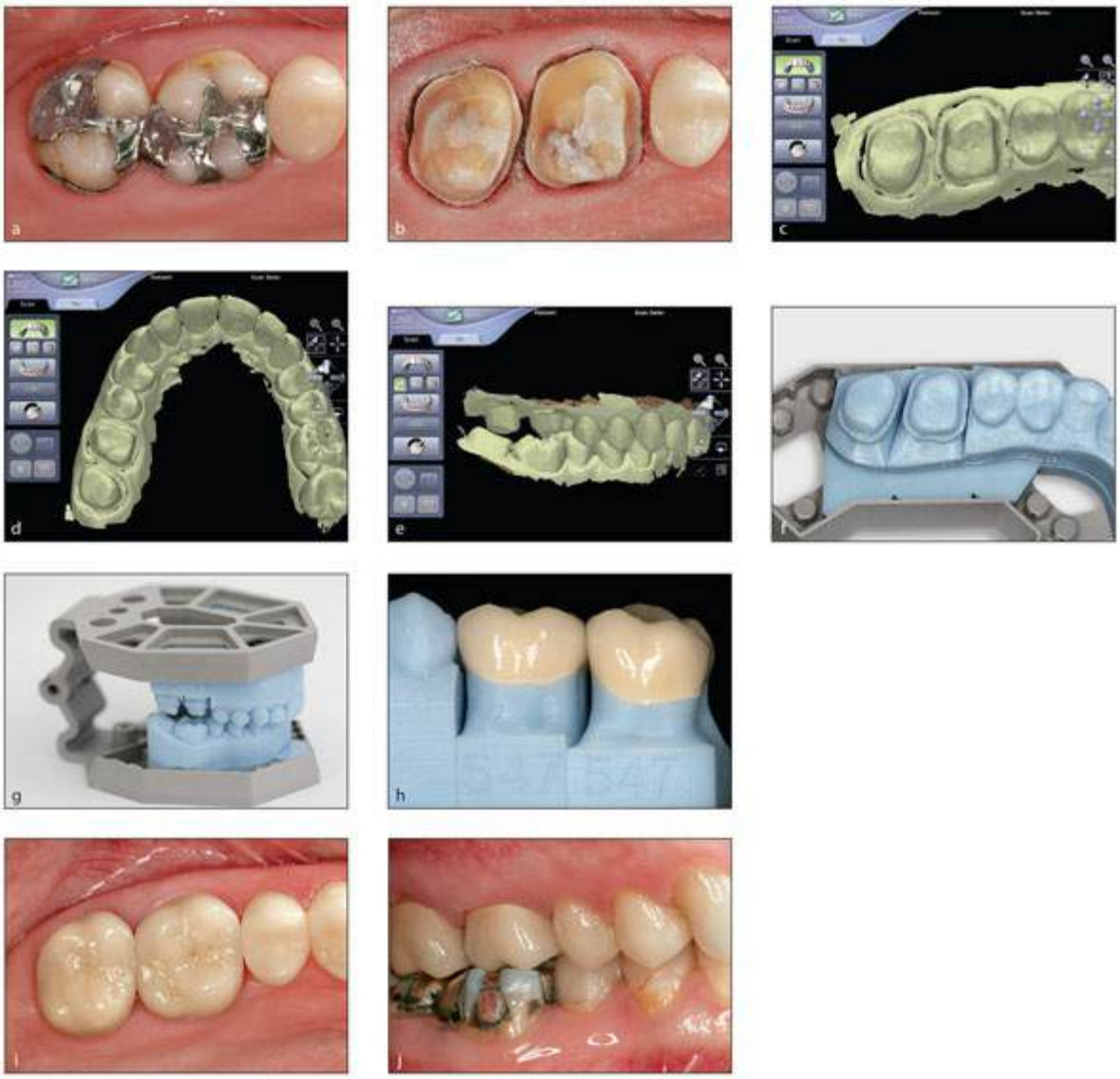


Fig 19-8 The digital impression procedure using the Lava Chairside Oral Scanner (3M ESPE). *(a)* Preoperative occlusal view of the maxillary right first and second molars. *(b)* Crown preparations following tissue retraction and isolation with a fine coating of titanium dioxide powder for digital imaging. *(c)* Digital recording of the crown preparations. *(d)* Digital recording of the full arch. *(e)* Digital models virtually articulated in centric occlusion with the buccal scan. *(f)* Crown preparations on the resin model. *(g)* Articulated resin models from the digital impressions. *(h)* Zirconia crowns (Lava, 3M ESPE) on the resin dies. *(i)* Occlusal view of the cemented Lava crowns. *(j)* Facial view of the cemented Lava crowns.



Fig 19-9 Lava Chairside Oral Scanner.



Fig 19-10 iTero system.

Chairside CAD/CAM systems can fabricate single-tooth ceramic or composite inlays, onlays, veneers, and crowns. The CEREC Connect system for the CEREC Acquisition Center (AC) unit (Sirona Dental) and the E4D Sky network for the E4D Dentist system (D4D Technologies) are recent upgrades that offer the opportunity to use the chairside systems as purely digital impression systems for cases desired to be fabricated in the dental laboratory[119] (Figs 19-11 and 19-12).



Fig 19-11 CEREC AC unit with the MC XL and MC milling chambers.



Fig 19-12 E4D Dentist system, consisting of the mobile Design Center and milling unit.

Computerized systems: Cameras

A critical first step in the CAD/CAM sequence is to accurately record the specific teeth and soft tissues to the computer software program. An intraoral camera or scanner becomes the key element for a digital impression. There are significant differences among the cameras and their manners of use in recording the digital impression.

The CEREC AC system has a blue light-emitting diode (LED) camera called the *Bluecam*. It records a series of single images that the software combines into the three-dimensional (3D) virtual model of the dentition. Although a single image is the minimum required to fabricate a restoration, additional images are routinely used to record adjacent teeth, mount the opposing virtual model, or aid in designing the restoration. A titanium dioxide powder is sprayed on the dentition and soft tissues to be recorded to create a uniform reflective surface. The shorter wavelength of the blue light compared with that of a laser allows for greater precision of the optical image.[120] It also uses a telecentric beam that records the tooth surface data from all visible surfaces in a single view.[121] The Bluecam can be used in a manual or automatic mode. The automatic-capture mode prevents recording of data while the camera is moving or shaking. This prevents the capture of blurred images that would be inaccurate.

The E4D Dentist IntraOral Digitizer is a red laser-light camera.[122] It records a series of overlapping single images, rather than a video recording, that the computer calculates into a virtual model. It does not require the use of a reflective powder unless scanning through a thin, transparent aspect of the preparation. In such cases, a

reflective liquid (E4D Accent) is applied to the surface of the tooth. The operator has the option of manual image capture or automatic image capture with Rapid Scan. A series of nine separate images are recorded from the occlusal, lingual, and facial views for a true 3D capture. Additional images are required to record adjacent teeth.[121] The software immediately indicates the accuracy and usefulness of each scanned image to ensure that all images are properly scanned.[121,122]

The iTero intraoral scanner uses a parallel confocal white light and red laser camera to record a series of images to create a 3D model.[122,123] The scanner emits a beam of light, and only an object at the correct focal length will reflect light back through the filtering device.[124] The scanner captures 100,000 points of laser light and focuses accurately to 300 focal depths spaced 50 μm apart.[121,125] The camera can be placed in contact with the teeth and does not need to be held stationary in space over the teeth. Scanning powder is not required for the camera to accurately record the surface of the teeth. The operator is prompted to record a series of five scans from the occlusal, facial, lingual, mesioproximal, and distoproximal angles of the prepared tooth as well as additional scans of adjacent teeth. The opposing dentition is scanned separately.[125] In addition, angled buccal and lingual views must be taken of the remaining teeth in the arch form.[122] Individual images may be retaken until adequate data is obtained.[121] A total of 15 to 30 scanned images may be required to record the preparation, opposing teeth, and occlusal relationships.[125] Ultimately, the software merges common data from all scans and proposes virtually articulated models.[122]

The Lava COS camera is the only video camera available for digital impressions. It uses "active wavefront sampling" to record up to 20 3D data sets per second and 2,400 3D data sets per arch. A light coating of a titanium dioxide powder is applied to the teeth and soft tissues to enhance the accuracy and efficiency of the scanning process. Three sensors record the clinical situation from varying perspectives and use proprietary image-processing algorithms to process the model.[124] Unlike a single-image camera, it captures 3D data in a video process and fabricates the virtual model in real-time on the computer monitor.[126] The virtual model can be switched between a 2D and 3D image as well as visualized with 3D glasses to verify aspects of the recorded models. Another feature of the Lava COS video capture is that the dentition can be recorded in strips or sections and the computer can assemble the strip scans into a real-time single 3D virtual model.[127] The operator has a field of view of approximately 10 mm by 13.5 mm with the camera. The camera has a 5- to 15-mm working depth from the surface being recorded to

record the dentition. This is sufficiently small to allow recording of the most distal posterior teeth. If the camera is held too close to or too far from the dentition, outside the focal distance, the video recording is automatically paused until the camera is brought back within the focal distance, and recording automatically continues. This prevents inaccurate data from being included in the scan.[122]

Digital impressions

All computerized dentistry systems rely on the ability to accurately record the dentition. Without a precise digital recording, an accurate restoration is not possible. The marginal fit and internal adaptation of the definitive restoration are directly related to the quality and accuracy of the recorded preparation. This concept is true for both conventional impressions and digital impressions. The definitive restoration can only be as accurate as the quality of the recorded data of the tooth.

There are some guidelines that hold true for all computerized dentistry systems. Digital recordings or scans are as sensitive to moisture contamination and soft tissue retraction problems as traditional impression materials. Moisture, such as saliva or blood, obscures the preparation and prevents an accurate recording of the tooth. Similarly, inadequate retraction of soft tissues may hide the marginal areas from view (Fig 19-13). Digital cameras can only record what is visible and isolated. Careful control of the scanning environment ensures an accurate digital file, which is essential to a well-fitting restoration.

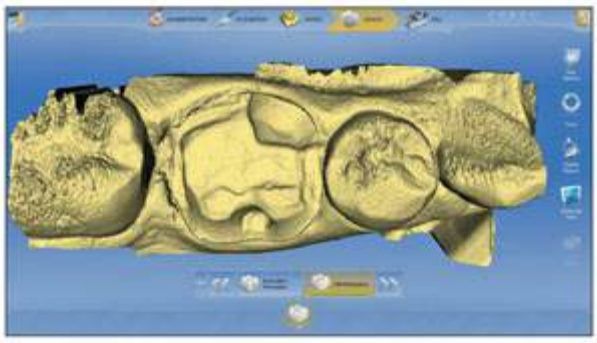


Fig 19-13 There are indistinct margins at the distolingual of the crown preparation for the mandibular first molar in this CEREC virtual model. This is a function of inadequate soft tissue retraction for good visibility of the margins.

Digital impressions offer the clinician excellent immediate feedback on the recorded preparation. The ability to greatly magnify the scanned data on the computer monitor is a significant advantage to ensure an accurate digital impression prior to transmitting the case to the dental laboratory for processing (Fig 19-14). For

conventional impressions, this level of magnification and critical evaluation may not be possible until the model has been poured and separated and the die trimmed.

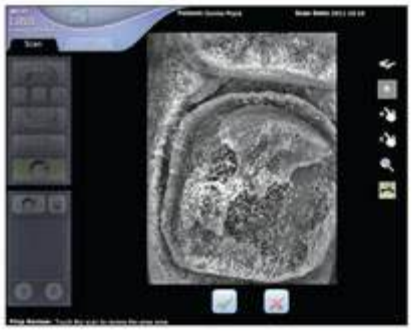


Fig 19-14 High-magnification alternative contrast view of a preparation margin with the Lava COS system allows for critical evaluation of the margins prior to transmitting the case to the dental laboratory.

Digital impressions also provide quantitative data on the preparation relative to the opposing dentition. This is especially helpful in ensuring adequate occlusal clearance for a case based on the specific planned restoration. The software program can provide quantitative data on the occlusal clearance between the models prior to submitting the case to the laboratory for processing (Fig 19-15). Immediate access to the high-magnification images allow for the correction of preparation deficiencies. This avoids discovering the deficiency in the stone models, which would require correction at a second appointment or the deficiency to be worked around by the dental laboratory technician.

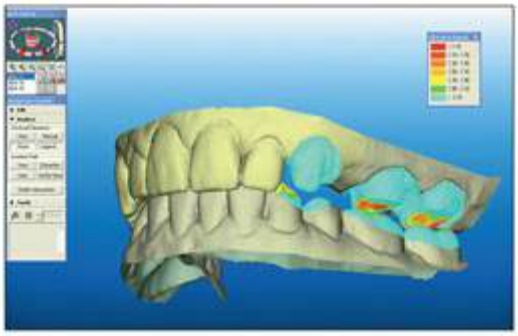


Fig 19-15 Quantitative information on the occlusal clearance is provided with a digital impression from the iTero system. The red-orange areas indicate less than 1.5-mm clearance from the opposing dentition.

Single-image cameras (CEREC AC, E4D, iTero) are line-of-sight cameras that record a series of single images of the dentition. The camera works in a manner similar to our eyes, in that we can see what is directly in our line of vision but not around corners. Each single image must overlap with one or more of the other recorded images to enable the computer to process the single images into a 3D

virtual model on the software program. Single-image cameras must capture images from a variety of angles to accurately record the dentition below the height of contour. Figure 19-16 shows the facial contour of a virtual model created from a series of single images made from the occlusal aspect of the teeth. Note the straight contours of the adjacent teeth below the height of contour. To accurately record the facial and lingual contours of the teeth, the camera must be angled past the height of contour with two or three images. Note that this is not a problem involving the preparation, as the camera can record all data within the path of insertion of the preparation in a single image.

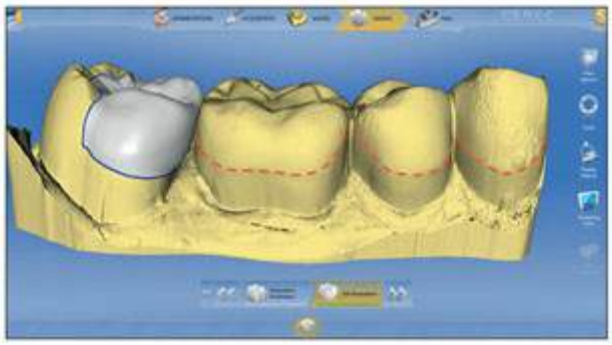


Fig 19-16 The *dotted lines* indicate the facial height of contour of the teeth. Note the flat emergence profile calculated by the software because the images recorded from the occlusal direction were not able to capture data hidden by the height of contour in this angle of recording.

Another important function in making a digital impression is the need to record opposing teeth and occlusal relationships for the planned restoration. Generally, two techniques are employed. One technique is to make a traditional bite registration (eg, with a polyvinyl siloxane bite registration material) of the dentition opposing the prepared teeth. The bite registration is scanned separately from the preparation model, and the software program matches the two models (Fig 19-17). The other technique is to digitally record the opposing model separately. The patient is guided into a closed centric position, and the dentition is scanned from the buccal view. This is referred to as a *buccal scan* (Fig 19-18). The buccal scan records the facial contours of the teeth and soft tissues. The software program uses the buccal scan to align the opposing digital models for accurate recording of the patient's occlusal relationship. No marketed system currently has the ability to digitally record functional lateral movements for mounting. Traditional lateral or protrusive functional registrations are required to mount the case on a semiadjustable articulator to simulate lateral guidance.

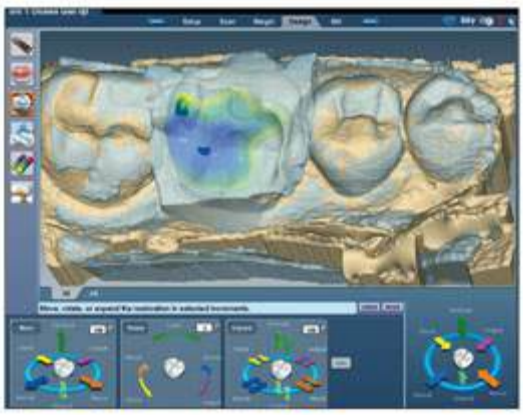


Fig 19-17 Superimposition of the bite registration model over the preparation model and restoration proposal to verify occlusal contacts with the E4D system.

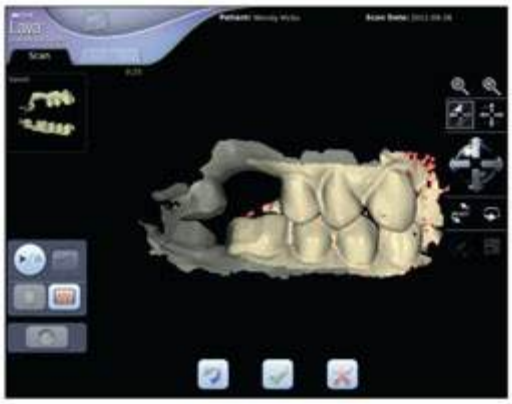


Fig 19-18 Buccal scan mounting of the scanned opposing models with the Lava COS system.

The undisputed key issue with digital impressions is accuracy. The use of conventional impression techniques and materials is the accepted technique for fabricating well-fitting restorations. Without an accurate impression, a well-fitting restoration is not possible regardless of the fabrication process. The starting point for considering digital impression systems is an expectation of at least equal accuracy to conventional impression techniques and materials. Digital impressions have been shown to be equally accurate to conventional impressions.[128,129] Several of the camera systems require the application of a titanium dioxide powder to the surfaces of the teeth and soft tissue to be scanned. The thin, uniform coating creates a uniformly reflective surface on the dentition for accurate data recording. A common misperception is that this thin powder layer limits the accuracy of the definitive restoration. However, even when used incorrectly, excess powder does not affect the marginal fit or internal adaptation any more than a die spacer does on laboratory-fabricated restorations using stone casts. Each of the digital systems has been shown to be capable of fabricating restorations with marginal fit and internal adaptation

equal to conventional fabrication techniques. CEREC has over 25 years of both laboratory and clinical research confirming the predictable outcomes possible with the chairside CAD/CAM system.[130–132] Lava COS has laboratory and clinical research confirming the digital impression technique as a consistently accurate restoration fabrication process.[124,128,129,133] In addition, the more recently introduced systems have laboratory studies documenting the marginal fit and internal adaptation of their restorations.[134,135]

Chairside CAD/CAM ceramic restorations

CAD/CAM technology for the fabrication of ceramic restorations has been a significant development in dentistry. The vast majority of CAD/CAM systems have been designed for use in the dental laboratory. InLab (Sirona Dental), Procera (Nobel Biocare), Lava (3M ESPE), and Cercon (Dentsply/Caulk) are but a few of the CAD/CAM systems dental laboratories are using in an effort to improve efficiency while utilizing modern ceramic materials. There are two currently marketed in-office or chair-side CAD/CAM systems: the CEREC AC driven by the Bluecam camera and the E4D Dentist system. The chairside CAD/CAM systems offer considerable time savings over conventional, laboratory-generated restorations because there is no need for a conventional impression, temporary restoration, or a second appointment; the definitive esthetic restoration is delivered at the same appointment at which the tooth is prepared.[136,137]

The CEREC system was initially marketed as the CEREC 1 system in Europe in 1985, and the first clinical trials took place in 1987.[138] It was introduced in the United States in 1989. Since then, the system has evolved through a series of hardware and software upgrades to reach its present form, the CEREC AC driven by the Bluecam.[139] The E4D Dentist system was introduced in 2008 with its DentaLogic software offering a true 3D virtual model.[140]

Hardware and software

The CEREC AC system consists of an acquisition unit and a milling chamber (see Fig 19-11). The acquisition unit contains a computer with the 3D design software, a liquid crystal display (LCD) monitor, and an intraoral camera for the optical imaging of the cavity preparation. The milling unit houses two motors. Each motor controls a cutting diamond for milling the definitive restoration from prefabricated blocks of restorative material. One motor has a step bur diamond that mills the entire

internal surface of the restoration. The other motor has a milling diamond that has a tapered cone shape and mills the entire external surface of the restoration. The milling unit also contains a water chamber for distilled water and lubricant additive that is used during the milling procedure to clean and lubricate the diamonds as they mill the restorative material. The acquisition and milling units communicate with each other via a radio transmitter.

The E4D Dentist system has an intraoral laser scanner, a mobile Design Center with DentaLogic software, and a separate milling chamber with a dedicated mill server computer (see Fig 19-12). The milling chamber has two opposing milling motors that can automatically change between three different milling diamonds depending on the specifics of the restoration to be milled. The milling chamber has a dedicated computer server to allow it to operate independently of the Design Center after the case design has been completed and transmitted to the job server.

Inlays/onlays

The tooth preparation will have a major impact on the fit accuracy of the milled restoration. Cavity preparations for CAD/CAM inlays and onlays are similar to those for conventional indirect ceramic inlays and onlays.[141–143] (Fig 19-19). The occlusal aspect of the preparation should be at least 1.5 mm thick in the central fissure and 2.0 mm thick over the cusps to provide adequate strength for the ceramic restoration. All cavosurface margins should be well defined and have a 90-degree butt-joint configuration. This will allow the camera to record an accurate image of the cavosurface margin while providing strength to the restoration. Bevels and knife-edge margins should not be used. The cavity walls of the proximal box should have at least 6 to 8 degrees of divergent taper.[144,145] All preparation floors should be smooth but not necessarily flat. Concavities created by the removal of carious tooth structure or prior restorations may be blocked out with a liner or base, but this is not required to obtain an accurate fit of the restoration.



Fig 19-19 Cavity preparation for a CAD/CAM chairside onlay.

Crowns

The tooth preparation is the same as described for other all-ceramic crowns (Fig 19-20). The computer graphic design process for a crown is similar to that used for fabrication of an onlay; the milling process is somewhat longer. Many clinicians choose to characterize the monochromatic shade of the ceramic blocks with shade modifiers and a glaze prior to adhesive cementation.[144]



Fig 19-20 Cavity preparation for a CAD/CAM chairside crown.

An additional crown preparation design has been suggested for endodontically treated teeth: the *endocrown* design. This design incorporates the core into the crown as a single restoration (Fig 19-21). This design significantly increases the surface area of the preparation available for adhesive cementation. It is particularly useful in teeth with short clinical crowns. This restoration has not been well evaluated clinically; however, preliminary reports from the University of Zurich, Switzerland, indicate that the cross-sectional area of the tooth may be a factor in retention and may be problematic for premolar teeth.[145]



Fig 19-21 Preparation of an endocrown for an endodontically treated tooth.

Computer-aided design

The process for fabricating a chairside CAD/CAM onlay with the CEREC AC system is illustrated in Fig 19-22. A well-isolated and dry field is necessary to ensure that the preparation is optically scanned with precision and accuracy. It is also critical to clearly isolate the gingival extent of the preparation from the adjacent soft tissues. Following completion of the tooth preparation (see Fig 19-22b), an optical impression is recorded. Because enamel, dentin, bases, and soft tissues do not reflect the Bluecam LED light equally well, the tooth must be coated with a

uniform reflective material before imaging. The powder should also coat all adjacent structures that may be viewed in the optical scan (see Fig 19-22c). The E4D Dentist IntraOral Digitizer only requires the use of the imaging powder in thinner, more transparent aspects of the preparation.



Fig 19-22*(a)* Preoperative view of the mandibular left first molar. *(b)* Onlay preparation for the first molar. *(c)* Preparation and adjacent teeth coated with titanium dioxide powder to record the digital impression. *(d)* Preparation virtual model is calculated for the design of the restoration. *(e)* Computer graphic design of the restoration. *(f)* The milling preview window shows the calculated 3D model of the restoration prior to milling. *(g)* Try-in of the milled Lava Ultimate (3M ESPE) onlay. *(h)* Lava Ultimate onlay is completed following adhesive cementation and polish. *(i)* Facial view of the completed Lava Ultimate onlay.

The camera images are recorded to the software program, where a virtual 3D model is rendered in real time (see Fig 19-22d). All portions of the preparation margins should be visible and distinct from adjacent structures. The camera head must be held motionless to record an accurate image without distortion of the preparation.

The design phase is initiated when the computer renders a virtual 3D model of the cavity preparation from the recorded optical image. The 3D capability of the software (both CEREC and E4D) allows for the preparation model to be rotated in an infinite number of angles and views for design of the restoration. Following identification of the cavity margin by tracing the margin line, the proposed restoration design is shown on the monitor (see Fig 19-22e). A number of editing

tools are available to customize the contours, anatomy, and occlusal relationship of the restoration.

Computer-assisted manufacture

When the milling function is activated, the software calculates a volume model of the restoration (see Fig 19-22f). Based on the calculated 3D model of the restoration, the computer then requests that the appropriately sized ceramic block be inserted into the milling chamber.

A number of restorative materials are available for use in both chairside CAD/CAM systems. All currently marketed milling materials are available for both systems with the exception of the Vident line of materials, which are restricted to use in the CEREC AC system.[146–148] The restorative material is fabricated in a block form and mounted on a milling stub that can be inserted into the milling chamber. Vita Mark II blocks (Vident) are made of a fine-grained, feldspathic porcelain that has a homogenous structure. The fine grain of the Vita Mark II porcelain is reported to increase hardness of the material but decrease wear of the opposing dentition.[149] Al-Hiyasat et al[150] reported that the Vita Mark II ceramic material was less abrasive and more resistant to wear than conventional aluminous and bonded ceramic materials. The material comes in nine Vita 3D-Master shades (Vident) as well as a bleach shade and an esthetic translucent line of shades. The TriLuxe Forte block (Vident), a multicolor block that simulates the variation in shade from cervical to incisal in a natural tooth, is also available, as well as a RealLife block (Vident) that simulates variation in shade from internal to external of a natural tooth. Improvements in the software allow for restoration designs to be properly positioned within the multicolor block prior to milling.

The first leucite-reinforced glass ceramic CAD/CAM block was ProCAD (Ivoclar Vivadent), introduced in 1998. It evolved to the current IPS Empress CAD and is a 35% to 45% leucite-reinforced glass ceramic similar to IPS Empress 1 but with a finer particle size of 1 to 5 μm. The excellent clinical success with Empress 1 is expected to be duplicated with the IPS Empress CAD material, as they are essentially the same restorations with different fabrication techniques. Both Vita Mark II and IPS Empress CAD materials can be characterized and glazed in a porcelain oven using stains that are applied chairside.

An innovative resin composite material is also available. Paradigm MZ100 (3M ESPE) is a resin composite with a filler composed of zirconia and silica.[151] The inorganic filler loading is 85% by weight, with an average particle size of 0.6 μm,

and the material is radiopaque. A study has reported a randomized, longitudinal clinical trial comparing Paradigm MZ100 inlays to Vita Mark II inlays over 3 years of clinical service.[152] A recent study reports the 10-year follow-up of this study as well.[153] The inlays were evaluated using modified US Public Health Service (USPHS) criteria on a yearly basis. The study determined that Paradigm inlays performed equally as well as Vita Mark II inlays at 10 years, with clinical advantages noted in inlay fracture resistance and color match.

A recent material addition for this type of restoration is Lava Ultimate (3M ESPE). Lava Ultimate contains a blend of nanosized silica and zirconia particles (1 to 100 nm) agglomerated into clusters and individual bonded nanosized particles embedded in a highly cross-linked polymer matrix, with approximately an 80% filler load. The material is purported to offer the ease of use of a composite material with a surface gloss and finish retention similar to that of ceramics. To date, no clinical studies have reported on the performance of this material.

Once the ceramic or composite block has been locked in the milling chamber, the unit is ready to mill the restoration. The milling diamonds move in coordination with the movement of the metallic mounting of the ceramic or composite block, which calibrates the position of the milling head. The ceramic or composite block, rotating as it goes, is fed uniformly through the milling process. Milling time is a function of the size and complexity of the restoration and may range from 10 to 20 minutes on average.

Try-in and cementation

Once the ceramic or composite restoration is recovered from the milling chamber, the restoration is tried in. Adjustments to the proximal contacts may have to be made for the restoration to seat completely if the preparation was too sharp or angular. The axiopulpal line angle and the gingival floor of the box are the places most likely for the restoration to require adjustment.[113] Limited to no occlusal adjustment should be done prior to cementation of the restoration; the ceramic is particularly fragile without the support of the luting cement. Should the fit be unacceptable in some area, the design can be reloaded from the computer hard drive and corrected, and a second restoration can be milled quickly.

The ceramic restoration should be etched with hydrofluoric acid and silanated prior to adhesive cementation similar to laboratory ceramic restorations. CAD/CAM composite and nanoceramic materials should be air abraded with CoJet rather than etched prior to silane treatment. Use of a conventional adhesive cementation

technique and a dual-cured resin cement is recommended.[137,138,154] Contouring and refinement of anatomy can be accomplished after cementation with various microfine diamonds.[155] Finishing and polishing are completed with abrasive rubber points, disks, and cups or brushes with diamond polishing paste as for other ceramic and composite restorations.

Advantages and disadvantages

The main advantages of chairside CAD/CAM technology are the saving of time, service to the patient, and utilization of optimal restorative materials. The technique affords the dentist the opportunity to prepare, design, and fabricate a ceramic restoration in a single appointment, without the need for conventional impressions, provisional restorations, or dental laboratory support. Chairside CAD/CAM technology offers excellent esthetics, durability, and at least short-term strengthening of the tooth. Roznowski et al[156] compared the fracture resistance of molars restored with various adhesive and nonadhesive restorations and reported that teeth restored with mesio-occlusodistal CEREC inlays were as strong as unprepared, unrestored teeth, while nonadhesive restorations weakened teeth.

The chairside CAD/CAM systems are not without their disadvantages. The systems rely on accurate data from the tooth preparation to fabricate the restoration. This requires developing skill in the use of the intraoral scanning device or camera as well as good soft tissue retraction and isolation of the dentition while using the camera. Because the restoration is fabricated without the benefit of a die or articulated casts, development of occlusal relationships may be problematic. The systems have several design options to overcome this apparent limitation. One technique provides the opportunity to record the pretreatment occlusal anatomy, or that of a diagnostic wax-up, and copy it to the occlusal surface of the new restoration design. The CEREC 4.0 program refers to this function as *Biogeneric Copy* and the E4D DentaLogic program refers to this as *Clone scanning*. The existing occlusal relationships can be maintained in the new restoration. An alternative technique available to the E4D system involves the use of a bite registration to record the occlusal surface of the opposing dentition. The surface data is recorded to the software and superimposed over the design of the new restoration. The CEREC 4.0 program uses a buccal scan technique. This involves recording images of the opposing dentition in a second model catalog and then virtually mounting the opposing models with a scan of the dentition from the facial aspect with the patient closed in centric occlusion position. These techniques offer a degree of control in

evaluating and developing occlusal relationships in the new restoration. It is not a "virtual articulator" in that it is a static recording and does not record interocclusal relationships in excursive mandibular movements. However, interocclusal relationships in maximum intercuspation position (centric occlusion) of the mandible can be accurately recorded to the system.[157] The milled restoration will require some adjustment after bonding to ensure the absence of interferences in excursive mandibular movements.

A significant training period is required to achieve proficiency with any CAD/CAM system. Participation in a 2-day initial training course is strongly recommended for the chair-side CAD/CAM systems prior to clinical application with the system. This is often followed with a second advanced course after 2 to 4 months of clinical experience. The initial cost of the system is also considerable; however, it is balanced against the minimal per-restoration cost of using the system as compared to laboratory-fabricated ceramic restorations. Generally, the actual cost of the materials for a chairside CAD/CAM restoration to the dentist is about 10% to 15% of the cost of a laboratory-fabricated ceramic restoration.

A major concern about chairside CAD/CAM restorations has been their precementation marginal fit. Several studies have evaluated marginal openings in CAD/CAM restorations. Krejci et al[158] reported precementation marginal openings with CEREC 1 restorations to be in the range of 125 to 175 μm. However, after cementation, they found that more than 90% of enamel-ceramic interfaces had "continuous margins." They found that the quality of the marginal adaptation immediately after cementation does not seem to depend on precementation marginal fit. Peters and Bieniek[159] reported results of a clinical study of 22 CEREC 1 inlays in which the average marginal fit was 121 μm with a range of 60 to 150 μm. Wilder[113] reported the average film thickness of resin cement at the occlusal cavosurface margin with CEREC 1 to be 89 ± 65 μm; the average thickness at the gingival margin was 105 ± 81 μm. By comparison, Christensen[160] reported that gold castings may be fabricated with marginal openings of less than 25 μm. Because of the relatively large precementation marginal gaps, the exposed resin cement was thought to be the weak link in the CEREC 1 restoration.

CEREC system hardware and software improvements have led to improvements in adaptation and marginal fit. Mörmann and Schug[161] evaluated the marginal fit of CAD/CAM inlays fabricated with the improved CEREC 2 unit using the COS 4.01 software program. They reported the overall mean interfacial margin width to be 56 ± 27 μm for CEREC 2 inlays and 84 ± 38 μm for CEREC 1 inlays. They also reported that the milling precision of the CEREC 2 unit was 2.4 times greater than

that of the CEREC 1 unit. Benz et al[162] also reported a mean occlusal interfacial width for CEREC 2 inlays of 48 ± 34 μm, a 35% improvement over the CEREC 1 inlays. Ellingsen and Fasbinder[132] compared the precementation fit of CEREC 2 crowns with that of CEREC 3 crowns. All areas of crown adaptation measured were significantly smaller for the CEREC 3 system compared with the CEREC 2 system except at the axial walls of the preparation. Cavosurface margin gaps of 47.5 ± 19.5 μm were reported for CEREC 3 crowns, and gaps of 97.0 ± 33.8 μm were reported for CEREC 2 crowns. The fit of CEREC 3 crowns is very nearly on par with that of gold castings.

Cook and Fasbinder[163] reported a study that evaluated the marginal fit and internal adaptation of CEREC crowns fabricated on virtual models made from the CEREC 3 infrared laser camera and CEREC AC blue LED camera. There was no significant difference in the marginal fit and internal adaptation of CEREC crowns fabricated with the infrared laser camera (Redcam) and the LED camera (Bluecam), with the average marginal gap for all groups measuring less than 67 ± 18 μm.

Postoperative sensitivity following adhesive restorative procedures is not an uncommon problem. This is often described by the patient as discomfort to cold or to biting pressure. However, clinical studies with chairside CAD/CAM restorations report a low rate of postoperative sensitivity. Sjögren et al[164] reported a higher rate of postoperative sensitivity than that reported in other chairside CAD/CAM studies, with 10 of 72 patients reporting postoperative sensitivity with Vita Mark I or II ceramic inlays. Heymann et al[165] reported no postoperative sensitivity at any recall interval in their 4-year clinical trial of CEREC ceramic inlays. Fasbinder et al[166] reported that 13% of 92 Vita Mark II onlays were rated slightly sensitive at 1 week and 4% at 2 weeks. All sensitivity was resolved by 1 month, and there was essentially no postoperative sensitivity throughout the remainder of the study. Another study by Fasbinder aet al[167] reported that 2 of 62 CEREC onlays cemented with a self-etching cement were sensitive at 1 week, with all sensitivity resolved at 3 weeks without treatment. No restoration was reported as sensitive over the remaining 3 years of the study. Similar postoperative sensitivity was reported in a 10-year randomized clinical trial of chairside CAD/CAM composite and porcelain inlays cemented with an etch-and-rinse concept and a dual-cured resin cement.[153] One of the 80 inlays was reported as sensitive at 1 week, and the sensitivity resolved by 2 weeks without treatment. No other postoperative sensitivity was reported over the remaining 10 years of the study.

The chairside CAD/CAM technique may play a role in minimizing postoperative

sensitivity. It is imperative to be able to isolate the cavity margins in order to record accurate images of the preparation to fabricate a well-fitting restoration. The ability to deliver the restorations in a single appointment prevents the potential for dentin contamination during the provisional phase. Also, the use of manufactured blocks of porcelain and resin composite significantly restrict the influence of polymerization shrinkage to the composite luting agent, and shrinkage concerns are limited to the thickness of the resin cement.

Longevity/clinical studies

The failure rate of Vita Mark II porcelain inlays made with the CEREC technique has been reported to be very low. Sjögren et al[168] reported 4 fractures among 66 Vita Mark II inlays over 5 years. Pallesen and Van Dijken[169] reported a single fractured Vita Mark II inlay among 16 pairs of CAD/CAM inlays over 8 years. Berg and Dérand[170] reported 3 fractures among 115 Vita Mark II inlays over 5 years. Zimmer et al[171] carried out a retrospective clinical study of 226 CEREC 1 restorations placed in a private practice and evaluated after 10 years. The Kaplan-Meier survival rate reported was 94.7% after 5 years and 85.7% after 10 years. Martin and Jedynakiewicz[172] performed a systematic review of clinical studies on intracoronal CEREC restorations. They reported a mean survival rate of 97.4% over a 4-year period. The primary reasons for failure were reported as fracture of the ceramic material, fracture of the supporting tooth, and postoperative sensitivity. Fracture of the ceramic material was generally a result of occlusal stress or insufficient material thickness. Otto and De Nisco[173] reported an 8% failure rate for ceramic inlays after 10 years of clinical service. Of the failures, 53% were caused by fractures of the ceramic material and 20% were because of tooth fracture. Hickel and Manhart[174] reviewed clinical studies in the dental literature during the 1990s and reported annual failure rates of posterior restorations in stress-bearing areas as 0% to 1⅛% for laboratory-fabricated resin composite inlays, 0% to 7.5% for laboratory-fabricated ceramic inlays, and 0% to 4.4% for CAD/CAM ceramic restorations. They also mentioned bulk fracture as a frequent cause of failure for ceramic inlays.

A significant number of long-term clinical studies have been published on chairside CAD/CAM restorations using all generations of the CEREC system. Zuellig-Singer and Bryant[175] reported results of a 3-year clinical evaluation of CEREC inlays with four different luting agents. One inlay fractured, and 94.6% of the restorations had continuous margins at 3 years. There was no significant

difference in continuous-margin rates among the different luting materials. The authors also reported less wear with the microfilled luting resins as compared to that of the hybrid composite and glass-ionomer luting materials. Heymann et al,[165] in the previously mentioned clinical study of CEREC 1 inlays, demonstrated no significant changes in the 50 CEREC inlays after 4 years of clinical service. There was no reported postoperative sensitivity, and there were no restoration fractures. Sjögren et al[176] reported a 10-year clinical evaluation of 66 CEREC inlays. They calculated an 11% failure rate after 10 years, with four inlay fractures, one tooth fracture, one restored tooth requiring endodontic treatment, and one inlay replaced because of postoperative symptoms. The survival rate at 10 years was estimated at 89%. They achieved a significantly improved survival rate for CAD/CAM-fabricated inlays cemented with a chemically cured resin composite cement as compared to those cemented with a dual-cured resin cement. Otto and De Nisco[173] reported a 10-year prospective study of 200 CEREC 1 inlays and onlays. The primary failure mechanisms of the 15 failed restorations included restoration fracture (eight cases) and tooth fracture (three cases). Only three of the teeth that had failed restorations required crowns; the remaining restorations were repaired or the initial restoration replaced with another CEREC inlay or onlay. The Kaplan-Meier survival rate was 90.4% at 10 years.

Fasbinder et al[153] reported a 10-year randomized clinical trial of 80 CEREC restorations comparing porcelain inlays with composite inlays. Both restorative materials were cemented with an etch-and-rinse technique and dual-cured resin cement. After 10 years, there was one fractured composite inlay and five fractured porcelain inlays. The composite inlays performed equally as well as the porcelain inlays in all USPHS categories, with significantly better color match and less bulk inlay fracture. In another study, Fasbinder et al[167] reported the results of a randomized, longitudinal clinical trial of 62 CEREC onlays. The leucite-reinforced porcelain and feldspathic porcelain onlays were cemented with a self-etching, self-adhesive resin cement. The onlay preparation extended at least one-half the intercuspal distance, and at least one cusp was not covered with the onlay. Two of the feldspathic porcelain onlays failed during the study. The leucite-reinforced and feldspathic onlays performed similarly well with a self-etching, self-adhesive cement at 3 years with no significant changes observed in any USPHS criteria.

Bindl and Mörmann[177] reported a study comparing CEREC crowns with reinforced ceramic-core crowns. Thirty-six anterior crowns in 24 patients were placed using equal numbers of CEREC and In-Ceram Spinell (Vident) crowns. The crowns were evaluated over a mean service time of 44.9 ± 10.3 months using

modified USPHS criteria. No significant difference in clinical performance between the two types of ceramic crowns was observed. Fasbinder et al[178] reported a longitudinal clinical trial of 62 lithium disilicate CEREC 3 crowns. No significant difference in the gingival index or plaque index was observed between the test crowns and the control teeth. The percentage of alpha scores for marginal adaptation, caries, and crown fracture was 100% for both groups of crowns at all recall intervals. No surface chipping or fracture was identified on any of the crowns after 2 years. Based on the studies cited above, the performance of the current generation of CAD/CAM-fabricated restorations is expected to be excellent, and the use of this technology is predicted to grow steadily.

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